

Rational Numbers – Aptitude Syllabus Notes

Language: Tanglish | **Purpose:** Aptitude / Competitive Exam Preparation

1. Rational Number – Basic Definition

Oru number-ai $\frac{p}{q}$ form-la represent panna mudinja, where p and q are integers and $q \neq 0$, antha number Rational Number.

Examples: $\frac{1}{2}$, $\frac{3}{4}$, $-\frac{5}{7}$, $\frac{8}{1}$, 0

Important: 0 rational number dhaan, because $0 = \frac{0}{1}$.

2. Why Denominator 0 Koodathu?

5/0 rational number illa. Division by zero is undefined. Denominator always non-zero-a irukkanum.

Golden rule: $q \neq 0$

3. Rational vs Irrational Numbers

Rational	Irrational
$\frac{p}{q}$ form-la represent panna mudiyum	$\frac{p}{q}$ form-la represent panna mudiyadhu
$\frac{1}{2}$, $\frac{3}{4}$, -5, 0.5	$\sqrt{2}$, $\sqrt{3}$, $\sqrt{5}$, π
Terminating decimal possible	Decimal non-terminating and non-recurring
Recurring decimal possible	

4. Number Hierarchy

Natural $\subset$ Whole $\subset$ Integer $\subset$ Rational

5 $\rightarrow$ Natural, Whole, Integer, Rational. -5 $\rightarrow$ Integer and Rational. $\frac{1}{2} \rightarrow$ Rational only (among these four categories).

5. Types of Rational Numbers

Positive: $\frac{3}{5}$, $\frac{7}{8}$, 5

Negative: $-\frac{3}{5}$, $-\frac{7}{8}$, -5

Zero: $0 = \frac{0}{1}$

6. Equivalent Rational Numbers

Numerator and denominator-ai same non-zero number-ala multiply/divide pannina value change aagathu.

Example: $\frac{1}{2} = \frac{2}{4} = \frac{3}{6} = \frac{4}{8}$

7. Standard Form

Standard form-la numerator and denominator-ku common factor 1 mattum irukkanum; denominator positive-a irukkanum.

Example: $\frac{12}{18} \rightarrow$ divide by 6 $\rightarrow \frac{2}{3}$

8. Negative Sign Rules

$$-3/5 = 3/-5 = -(3/5)$$

$$(-3)/(-5) = +3/5$$

One negative → negative; Two negatives → positive.

9. Comparing Rational Numbers

a/b and c/d compare panna cross multiplication use pannalaam: compare **$a \times d$** with **$b \times c$** .

Example: $3/5$ vs $4/7 \rightarrow 3 \times 7 = 21, 4 \times 5 = 20 \rightarrow$ therefore $3/5 > 4/7$.

10. Addition of Rational Numbers

Same denominator: $2/5 + 1/5 = 3/5$

Different denominator: $1/2 + 1/3 \rightarrow \text{LCM } 6 \rightarrow 3/6 + 2/6 = 5/6$

11. Subtraction

$3/4 - 1/6 \rightarrow \text{LCM } 12 \rightarrow 9/12 - 2/12 = 7/12$

12. Multiplication

$a/b \times c/d = (a \times c)/(b \times d)$.

Example: $2/3 \times 4/5 = 8/15$.

Cancellation trick: Multiplication-ku munnaadi numerator/denominator cross-cancel pannina calculation easy aagum.

13. Division

Division means reciprocal multiply. **Keep $\rightarrow$ Change $\rightarrow$ Flip.**

$2/3 \div 4/5 = 2/3 \times 5/4 = 10/12 = 5/6$.

Formula: $a/b \div c/d = a/b \times d/c$

14. Decimal $\rightarrow$ Rational Number

$0.5 = 5/10 = 1/2$

$0.25 = 25/100 = 1/4$

$0.125 = 125/1000 = 1/8$

15. Terminating Decimal

Decimal finite-a stop aagum. Examples: $1/2 = 0.5$, $1/4 = 0.25$, $1/8 = 0.125$.

16. Recurring Decimal

Digits repeat aagite irundha recurring decimal. Example: $1/3 = 0.3333... = 0.3\overline{3}$. $2/7 = 0.285714285714...$.

Recurring decimal is rational.

17. Terminating Decimal – Important Exam Rule

Fraction lowest form-la irukkumbodhu denominator-la **2 and/or 5** prime factors mattum irundha decimal terminating.

3/4: $4 = 2^2 \rightarrow$ terminating.

7/20: $20 = 2^2 \times 5 \rightarrow$ terminating.

5/12: $12 = 2^2 \times 3 \rightarrow$ 3 irukku, so non-terminating recurring.

18. Recurring Decimal → Fraction

Example: $x = 0.3333\dots$

$$10x = 3.3333\dots$$

$$10x - x = 3$$

$$9x = 3 \rightarrow x = \frac{1}{3}$$

19. Rational Number Between Two Numbers

2 and 3-ku between one rational number: $\frac{5}{2}$.

$$2 < \frac{5}{2} < 3.$$

Important property: Any two distinct rational numbers-ku between **infinitely many rational numbers** irukkum.

20. Closure Property

Rational numbers addition, subtraction, multiplication-la closed. Division-layum closed, but divisor zero-a irukka koodathu.

Operation	Closed?
Addition	Yes
Subtraction	Yes
Multiplication	Yes
Division	Yes, divisor $\neq 0$

21. Commutative Property

Addition: $a+b = b+a \rightarrow \text{Yes}$

Multiplication: $ab = ba \rightarrow \text{Yes}$

Subtraction: Not commutative

Division: Not commutative

22. Associative Property

Addition: $(a+b)+c = a+(b+c) \rightarrow \text{Yes}$

Multiplication: $(ab)c = a(bc) \rightarrow \text{Yes}$

Subtraction / Division: No

23. Identity Elements

Additive identity = 0: $a + 0 = a$

Multiplicative identity = 1: $a \times 1 = a$

24. Reciprocal

a/b -oda reciprocal = b/a . Example: reciprocal of $\frac{3}{7}$ is $\frac{7}{3}$.

$$\frac{3}{7} \times \frac{7}{3} = 1.$$

0-ku reciprocal illa.

25. Additive Inverse

a-oda additive inverse = $-a$. Example: $3/5$ -oda additive inverse = $-3/5$.
 $3/5 + (-3/5) = 0$.

26. Multiplicative Inverse

a/b -oda multiplicative inverse = b/a . Product = 1. **0-ku multiplicative inverse illa.**

27. Aptitude Quick Tricks

1. Every integer is a rational number.
2. Terminating decimal is rational.
3. Recurring decimal is rational.
4. $\sqrt{\text{(perfect square)}}$ is rational: $\sqrt{25} = 5$.
5. $\sqrt{2}, \sqrt{3}, \sqrt{5}$ are irrational.
6. π is irrational.
7. Denominator 0 $\rightarrow$ not a rational number.
8. Any two distinct rational numbers-ku between infinitely many rational numbers.
9. Terminating decimal test: lowest-form denominator has only 2 and/or 5.
10. Division $\rightarrow$ Keep, Change, Flip.

28. Final Aptitude Checklist

- Rational number definition p/q , $q \neq 0$
- Rational vs irrational
- Natural / Whole / Integer / Rational hierarchy
- Equivalent fractions
- Standard form
- Positive / negative rational numbers
- Cross multiplication comparison
- Addition and subtraction
- Multiplication and cancellation
- Division and reciprocal
- Decimal to fraction
- Terminating decimal rule (2 and 5)
- Recurring decimal
- Recurring decimal to fraction
- Rational number between two numbers
- Closure, commutative, associative properties
- Identity and inverse

Core Formula: Rational Number = p/q , where p and q are integers and $q \neq 0$.